

## QUESTION 1 [20 marks]

- a. **(10 marks)** A voltmeter is used to measure  $V_0$  in the circuit in Figure 1. The voltmeter model consists of an ideal voltmeter in parallel with a  $250\text{ k}\Omega$  resistor. Let  $V_s = 30\text{ V}$ ,  $R_s = 10\text{ }\Omega$ ,  $R_1 = 20\text{ k}\Omega$  and  $R_2 = 15\text{ k}\Omega$ .
- (3 marks)** Output voltage ( $V_0$ ) in the circuit without voltmeter.
  - (4 marks)** Output voltage ( $V_0$ ) in the circuit with voltmeter.
  - (3 marks)** Equivalent resistance ( $R_{eq}$ ) of the circuit without voltmeter when the voltage source is turned off ( $V_s = 0\text{ V}$ ).

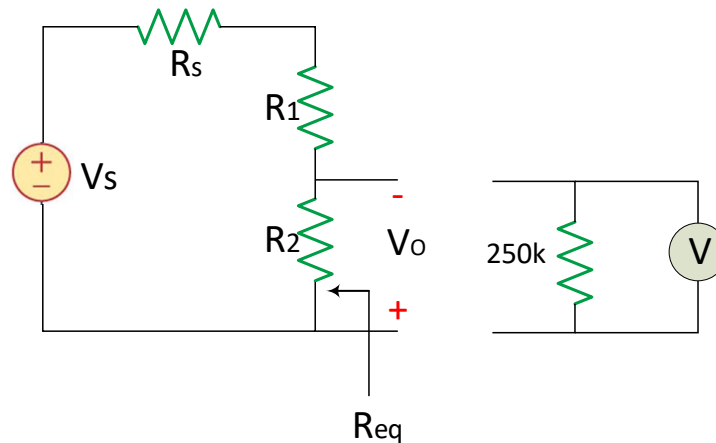


Figure 1

- b. **(10 marks)** The circuit in Figure 2 is used to control the light intensity of a portable lamp. The circuit consists of a  $12\text{ V}$  battery, a variable resistor and a light globe. The resistance of the light globe filament is  $4\text{ }\Omega$  and can be assumed to be independent of its temperature. The circuit also has a  $2\text{-A}$  fuse with negligible resistance.
- (5 marks)** If we want that the power dissipated in the light globe is  $9\text{ W}$ , what is the required value of the variable resistance?
  - (5 marks)** What value of variable resistance would make the fuse burn out? Note that a fuse burns out when the circuit draws more current than its amperage rating.

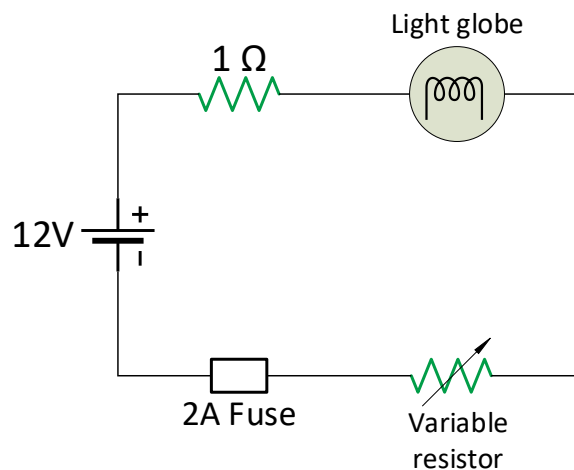


Figure 2

## QUESTION 2 [20 marks]

a. (10 marks) In the circuit shown in Figure 3,

- (7 marks) Find the voltage  $v_{ab}$ .
- (3 marks) Calculate the power in the dependent source.

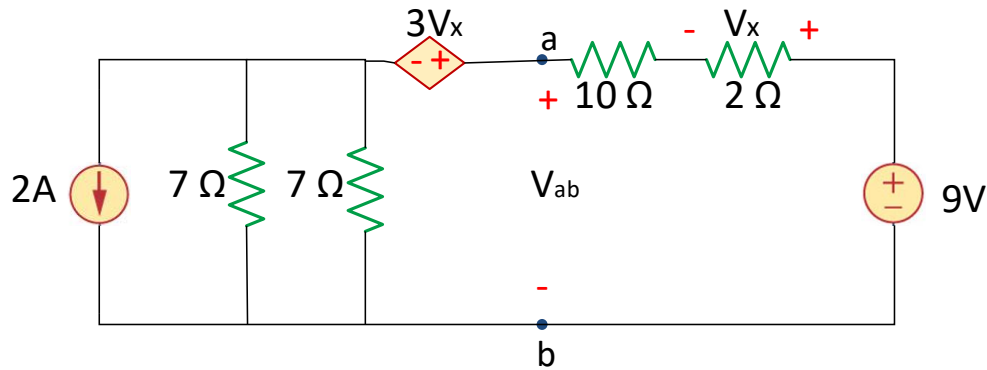


Figure 3

b. (10 marks) For the circuit shown in Figure 4, apply nodal analysis and write down the nodal equations using the labels provided.

Note: Simplify the equations in the form  $av_1 + bv_2 + cv_3 = d$ , where coefficients  $a$ ,  $b$ ,  $c$  and  $d$  can be positive or negative numbers, but **DO NOT** solve them.

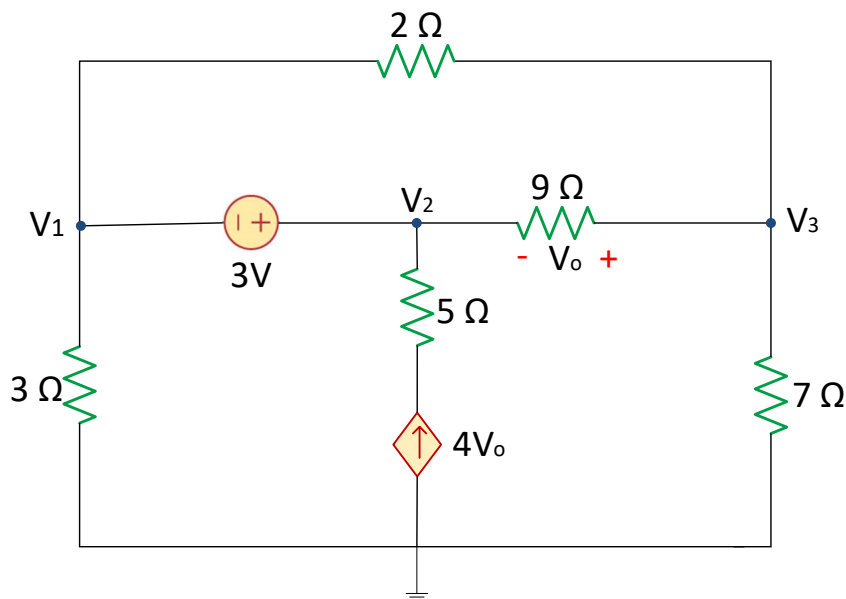


Figure 4

### QUESTION 3 [12 marks]

The circuit shown in Figure 5 is being used to illuminate a mine. Generator 1 is at the entrance of the mine and generator 2 is at a campsite. The lighting system is made from 20 light bulbs in parallel, where each light bulb has a nominal resistance of  $250\Omega$ . The resistance of the wiring within the lighting system is negligible and the resistance of the cables that connect the generators to the lighting system are as marked in the figure.

- (2 marks)** Find the value of the equivalent resistance of the lighting system.
- (10 marks)** Use mesh analysis to calculate the power dissipated by the lighting system.

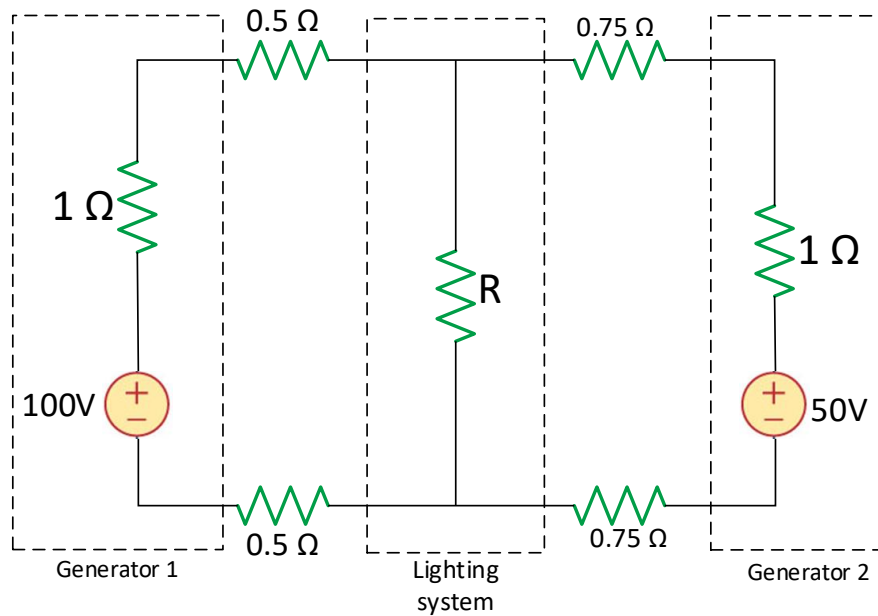


Figure 5

#### QUESTION 4 [28 marks]

- a. **(12 marks)** For the circuit shown in Figure 6, use a succession of source transformations (only source transformations and resistor/source simplification) to find an equivalent circuit consisting of a single voltage source and a single resistor, as seen from terminals  $a$ - $b$ . Then, determine the Thevenin voltage,  $V_{Th}$ , and Thevenin resistance,  $R_{Th}$ , at the terminals  $a$ - $b$ .

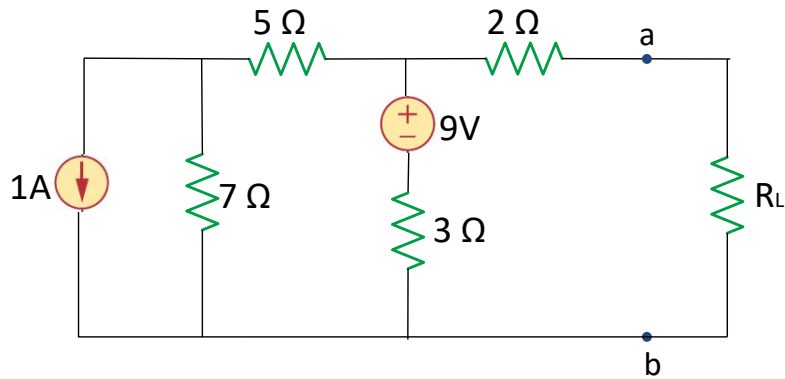


Figure 6

- b. **(16 marks)** The circuit in Figure 7 corresponds to a solar-powered lamp post. The solar panel charges the battery during the day. The battery is used to turn the light ON during the night.
- (9 marks)** Use the superposition principle to find the Thevenin voltage at the lamp post terminals (terminals  $a$ - $b$ ) during the day (when the solar panel is generating power).
  - (5 marks)** Calculate the lamp bulb resistance that will ensure the maximum transfer of power from the circuit to the lamp bulb.
  - (2 marks)** Find the maximum power that can be delivered to the lamp bulb.

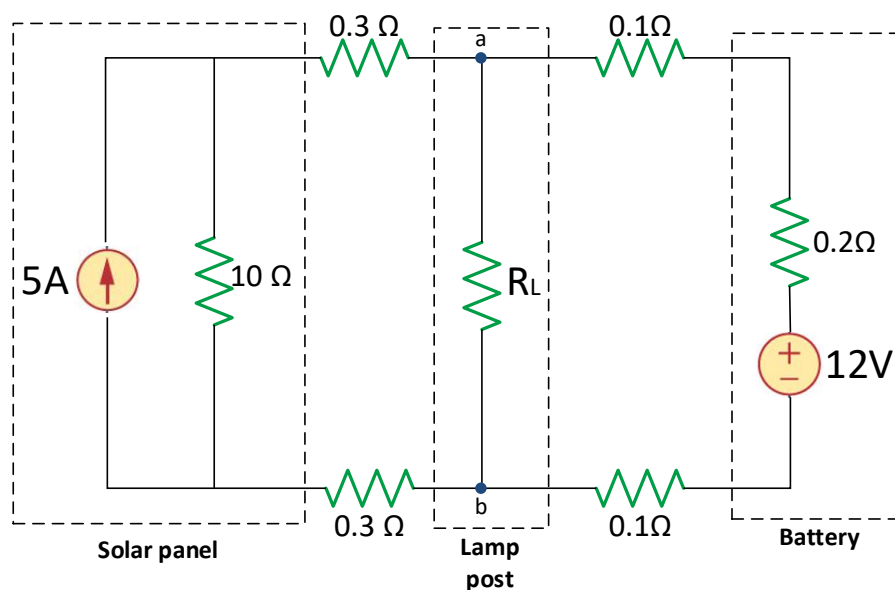


Figure 7

### QUESTION 5 [20 marks]

The switch in the circuit shown in Figure 8 was open for a long time. It closed at  $t = 0$ s and after remaining closed for 5 ms, it opened again.

- i. **(10 marks)** Calculate  $v_c(t)$  for  $0 \leq t < 5$  ms.
- ii. **(6 marks)** Calculate  $v_c(t)$  for  $t \geq 5$  ms.
- iii. **(4 marks)** Calculate the value of  $i_c$  at  $t = 10$  ms.

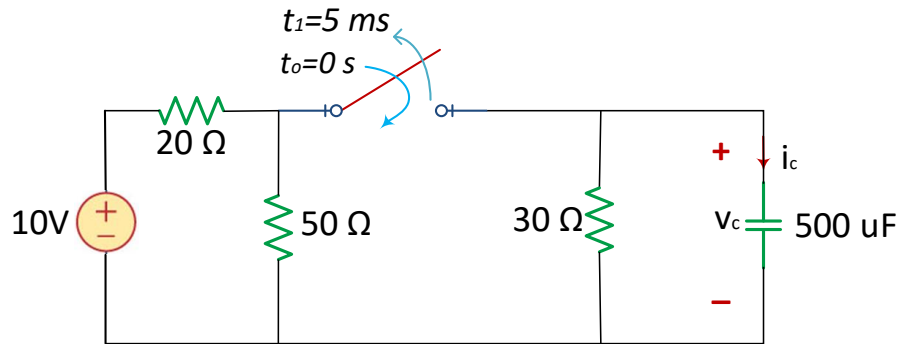


Figure 8